SFARI Screening Report

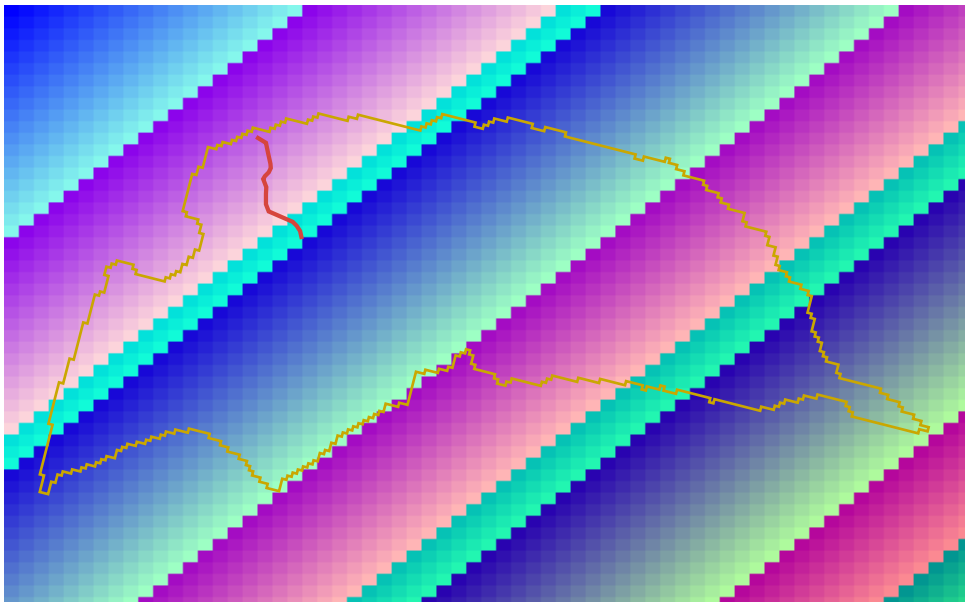
(unnamed stream)

Scenario: Restored riffle

Banks graded & replanted <2 yr>.

Coordinates	43.685784, -72.236713
COMID / HUC8	9327042 / 01080104
Drainage area	1.0005 km2
HR reach watershed area	1.0005 km2
Reach length	1000.0 ft
Ecosystem Condition Index	0.04

Watershed



Function scores (0-15)

Category	Function	Score	Condition
Hydrology	Catchment hydrology	11	F
Hydrology	Surface water storage	4	NF
Hydrology	Reach inflow		
Hydrology	Streamflow regime		
Hydraulics	Low flow and baseflow dynamics		
Hydraulics	High flow dynamics		
Hydraulics	Floodplain connectivity		
Hydraulics	Hyporheic connectivity		
Geomorphology	Channel evolution		
Geomorphology	Channel and floodplain dynamics		
Geomorphology	Sediment continuity		
Geomorphology	Bed composition and bedform dynamics		
Physicochemistry	Light and thermal regime		
Physicochemistry	Carbon processing		
Physicochemistry	Nutrient cycling		
Physicochemistry	Water and soil quality		
Biology	Habitat provision		
Biology	Population support		
Biology	Community dynamics		
Biology	Watershed connectivity		

Outcome sub-indices

Physical	Chemical	Biological	ECI
0.09	0.02	0.02	0.04

Metric evidence & Likert scores

Function	Metric	Likert	Pulled evidence
Catchment hydrology	Impervious surface area	A	6.0% impervious, 2.1% agricultural land (HR reach watershed)†
Catchment hydrology	Road density	A	1.71 km/km2 road density (HR reach watershed)
Catchment hydrology	Land use change		impervious 5.5% (2001) → 6.0% (2021), Δ +0.5 pts (HR reach watershed)
Catchment hydrology	Impoundments		0 NID dam(s) in the watershed, 0 acre-ft normal storage
Surface water storage	Wetland Coverage		13.2% wetland (HR reach watershed)
Surface water storage	Floodplain Water Retention		
Surface water storage	In-channel Ponding / Beaver		
Surface water storage	Off-channel Storage		
Reach inflow	Concentrated flow inputs		1 road-stream crossing(s), 1.00 crossings/km2 (HR reach watershed)
Reach inflow	Tributary condition and impact		
Reach inflow	Local runoff diversions		
Reach inflow	Road/highway drainage		
Streamflow regime	Flow permanence		0.0% zero-flow days at gage 01140817 (1078 d)
Streamflow regime	Flow statistics		Q10 0.2 / Q50 0.1 / Q90 0.07 cfs (gage 01140817)
Streamflow regime	Channel natural flow regime		baseflow ratio Q90/Q50 = 0.583 (gage 01140817), upstream normal storage 0 m3/km2 (HR reach watershed)
Streamflow regime	Artificial structures and inputs		0 dam(s) within ~1 mi (also check NLD levees)
Low flow and baseflow dynamics	Low flow velocity		
Low flow and baseflow dynamics	Low flow depth		
Low flow and baseflow dynamics	Longitudinal connectivity		
Low flow and baseflow dynamics	Lateral connectivity		
High flow dynamics	Overbank Flow Frequency		bank height ratio 1.68 (reach median, 3DEP sections)
High flow dynamics	Peak Flow Capacity (Morphological Check)		regional bankfull width 3.1 m, entrenchment ratio 1.54 (reach median, 3DEP sections)
High flow dynamics	Peak Flow Capacity (Velocity/ Shear Stress)		
High flow dynamics	Bed Mobilization Frequency		
Floodplain connectivity	Floodplain Complexity		
Floodplain connectivity	Entrenchment (ER)		
Floodplain connectivity	Channel Condition		
Floodplain connectivity	Lateral Floodplain Inundation		4.9 ac of NWI wetland within 150 m of the reach (18.4% of the strip), 0.25 ac of it riverine · entrenchment ratio 1.54 (reach median, 3DEP sections)
Hyporheic connectivity	Channel Complexity for Exchange		
Hyporheic connectivity	Floodplain Permeability		
Hyporheic connectivity	Bed Surface Grain Size		
Hyporheic connectivity	Visible Hyporheic Indicators		

Function	Metric	Likert	Pulled evidence
Channel evolution	Channel evolution stage		
Channel evolution	Incision trend (headcuts)		
Channel evolution	Widening trend		
Channel evolution	Recovery indicators		
Channel and floodplain dynamics	Bank Erosion Potential		
Channel and floodplain dynamics	Bank Migration and Meander		
Channel and floodplain dynamics	Sinuosity		
Channel and floodplain dynamics	Channel pattern		sinuosity 1.061
Sediment continuity	Sediment Deposition Patterns		
Sediment continuity	Channel Degradation (Incision)		
Sediment continuity	Fine Sediment Balance		
Sediment continuity	Transport Capacity		channel slope 0.0100 m/m, watershed agriculture 2% (HR reach watershed), soil K 0.31 (HR reach watershed)
Bed composition and bedform dynamics	Large Wood Frequency and Diversity		
Bed composition and bedform dynamics	Riparian Wood Recruitment		
Bed composition and bedform dynamics	Substrate Composition		
Bed composition and bedform dynamics	Bedform Diversity		
Light and thermal regime	Riparian Canopy Cover		48.4% forest in the 100 m riparian buffer
Light and thermal regime	Stream Temperature (rapid)		
Light and thermal regime	Algal Growth (light-limited)		
Light and thermal regime	Thermal Refugia		
Carbon processing	CPOM Retention		
Carbon processing	Detritus Decomposition Rate, Shredder/ Detritivore Presence		
Carbon processing	Riparian Corridor Width and Quality		73.1% natural vegetation in the 100 m riparian buffer (forest + shrub + grassland + wetland)
Carbon processing	Algal and Primary Production		
Nutrient cycling	Visible Algal Indications (N/P)		
Nutrient cycling	N/P Concentrations		observed median: TN 0.3 mg/L, TP 0.011 mg/L
Nutrient cycling	Vegetated Riparian Corridor Width		73.1% natural vegetation in the 100 m riparian buffer (forest + shrub + grassland + wetland)
Nutrient cycling	Relative Denitrification Potential		
Water and soil quality	Water Clarity / Turbidity		
Water and soil quality	Dissolved Oxygen (rapid)		
Water and soil quality	Pollutants		
Water and soil quality	pH & Specific Conductivity (rapid)		
Habitat provision	In-Stream Habitat Complexity		
Habitat provision	Overhanging Vegetation		

Function	Metric	Likert	Pulled evidence
Habitat provision	Aquatic Invertebrate Habitat		
Habitat provision	Lateral and Off-Channel Habitats		
Population support	Fish habitat		
Population support	Aquatic Invertebrate Community		
Population support	Fish Presence		
Population support	Amphibian and Crayfish Presence		
Community dynamics	Native and non-native species		
Community dynamics	Species richness, composition and abundance		
Community dynamics	Riparian communities		73.1% natural vegetation in the 100 m riparian buffer (forest + shrub + grassland + wetland)
Community dynamics	Keystone species		
Watershed connectivity	Upstream and Downstream Barriers		0 dam/barrier(s) within ~1 mi · CONN integrity 0.97 (StreamCat, COMID 9327042)
Watershed connectivity	Culvert and Road-Crossing Passability		
Watershed connectivity	Lateral Connectivity for Riparian Fauna		
Watershed connectivity	Dewatered or Intermittent Segments		NHD flow permanence: intermittent · HYD integrity 0.98 (StreamCat, COMID 9327042) · base-flow index 48% (HR reach watershed)

† Marked desktop evidence comes from the nearest StreamCat reach downstream.

Desktop evidence supports scoring; the assessor assigns the Likert and 0-15 function scores. Likert thresholds are national defaults. Calibrate regionally.

Site photos

Catchment hydrology: Impervious surface area

